

Types of events on one die roll

simple: $\{3\}$

1 2 3 4 5 6

exactly one outcome

compound: $\{5, 6\}$

1 2 3 4 5 6

two or more outcomes

mutually exclusive and exhaustive

1 2 3 4 5 6

$\{1, 2, 3\}$ and $\{4, 5, 6\}$: never together,
together they cover S

overlapping, not exhaustive

1 2 3 4 5 6

$\{1, 2, 3\}$ and $\{3, 4, 5\}$: share 3,
6 is in neither

impossible: $\emptyset$

no outcome at all

“roll a 7”: probability 0

sure: S

1 2 3 4 5 6

“roll 6 or less”: probability 1